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Clay Mathematics Institute
Scientific Advisory Board

Dear Members of the Board,

I enclose an article, *Constructive Existence and Qualitative Uniqueness of Turbulent Navier–Stokes Solutions*, which I have posted to arXiv. This letter says briefly what it contains.

Alternative (A) of the Millennium Problem asks, for all admissible data and with no forcing, for a solution of the incompressible Navier–Stokes equations that exists and is smooth for all time. What the article presents is a constructive solution *directed to* (A): for data given in advance — a domain in three dimensions, initial data, forcing and boundary conditions, at high Reynolds number — it exhibits turbulent solutions and establishes that their mean-value outputs, such as drag and lift, are determined to a stated precision. I do not claim this discharges (A) as that alternative is formulated, and the difference is stated plainly in the article: what is delivered is constructive existence and qualitative uniqueness, and not smoothness, because the solutions that occur are not smooth. Their gradients grow like $\nu^{-1/2}$ while their velocities stay bounded.

The method is the automated solution of differential equations, implemented in the FEniCS software: a finite element discretisation of the given equation on a given mesh, with output error control from the residual weighted by the solution of a dual problem, feeding back to optimisation of the mesh. The error in each output is thus bounded by a quantity the computation itself produces.

Fefferman writes that the four alternatives are offered “to give reasonable leeway to solvers while retaining the heart of the problem”. The Institute’s own description of the problem locates that heart in turbulence. I have taken the leeway and used it to reach turbulence, and I offer the result for what it is worth.

On prior publication. The material is not new. The Clay problem was first addressed on these lines in Chapter 14 of *Computational Turbulent Incompressible Flow*, a book by Johan Hoffman and myself published by Springer in 2007 as volume 4 of *Applied Mathematics: Body and Soul*, and the underlying computational record appears in refereed journals from 2006 onward, cited in the article. The requirement of publication in a refereed journal, with two years elapsed, is thus met by a margin approaching two decades. The remaining condition, that the work enjoy general acceptance in the community, is not mine to declare.

On the occasion. What is new is that a solution of alternative (C) has been announced, by methods themselves worth attention: a large optimisation fitting parameters to data by forward and backward propagation, constructing an object rather than asserting that one exists. The article observes that the method above is a restricted form of the same thing, and asks what it returns when directed at (A). It also records a scaling test showing that a singular solution of the kind constructed for (C) does not survive the limit of small viscosity in which fluid mechanics lies.

One interest should be declared: the industrial simulations cited in the article are the work of Icarus Digital Math, of which I am a co-founder. The comparisons themselves are set and held by the AIAA High Lift Prediction Workshop and the high-order CFD workshops, where predictions are submitted before the measurements are disclosed.

Yours faithfully,

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